

# gRPC MACH Architecture

## TCO Networking Costs Calculator

Point-to-Point Integration Pattern · Protocol Buffer IDL · AWS Egress Pricing

✔ File 'complexBiDirectionalStreamingRPCPattern.proto' uploaded successfully!



### Phase 4 — TCO Networking Costs Report

Full cost breakdown including architecture tactics, unit economics, and stakeholder summary.

SERVICE UNDER ANALYSIS

order-m-s

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🔊 BUC4

⚡ 950 base RPS

👥 500 End Users

📁 —

#### APPLIED ARCHITECTURE TACTICS

None selected — base configuration

| MESSAGE<br>TYPE | PROTO<br>SIZE<br>(BYTES) | EFFECTIVE<br>SIZE<br>(BYTES) | REQUESTS<br>/ MONTH | THROUGHPUT<br>(GB/MONTH) | TRANSFER COST<br>(USD/MONTH) |
|-----------------|--------------------------|------------------------------|---------------------|--------------------------|------------------------------|
| Request         | 8049                     | 8049                         | 2B                  | 18,458.6808<br>GB        | —                            |
| Response        | 11386                    | 11386                        | 2B                  | 26,111.3852 GB           | \$ 2,270.67                  |

#### 🗉 Total Cost of Ownership (TCO) Breakdown

| Cost Component                 | DR / BC    | Monthly<br>Cost | % of<br>TCO |
|--------------------------------|------------|-----------------|-------------|
| AWS Egress (Response Transfer) | Networking | \$2270.67       | 86.6%       |

| Cost Component                                                                                               | DR / BC           | Monthly Cost | % of TCO |
|--------------------------------------------------------------------------------------------------------------|-------------------|--------------|----------|
| Cloud Infrastructure<br>Database/Backup \$4.60   Security Services \$152.00   Containerized Cluster \$193.45 | Cloud + Container | \$350.05     | 13.4%    |
| Total Cost of Ownership (TCO)                                                                                |                   | \$2620.72/mo | 100.0%   |

\* % of TCO is calculated against the sum of ALL cost rows shown above (must total 100%). DR = Disaster Recovery | BC = Business Continuity.

 Unit Economics

TOTAL MONTHLY TCO (NETWORKING + CLOUD INFRA)

\$2620.72

\$2270.67 egress + \$345.45 cloud infra

 Per-Microservice Cost Metrics

TCO / MICROSERVICE / MONTH

\$2620.72

This service · full TCO

COST / REQUEST

\$0.00000106

2.462.400.000 req/mo

ANNUAL PROJECTION

\$31448.64

12× monthly TCO

 Per-End-User Cost Metrics (500 end users)

TCO / END USER / MONTH

\$5.2414

\$2620.72 ÷ 500 users

TCO / END USER / DAY

\$0.174715

Monthly ÷ 30

BUSINESS USE CASE

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Pattern cost attribution

 Revenue, ARPU & ROI Analysis

**Revenue model:** The value entered in Phase 1 is the *expected monthly revenue per end user* for this service. Total monthly revenue = that value × number of end users. **ARPU** = revenue per user per month. **ROI** = (total revenue – TCO) ÷ TCO. **Break-even users** = minimum end users needed to cover full TCO at current ARPU. Net Margin per user = ARPU – TCO per user. Present to Finance and Leadership to frame architecture cost as a revenue-generating investment.

TOTAL MONTHLY REVENUE

ARPU / MONTH

NET MARGIN / USER / MO

**\$5500**

\$11.00/user × 500 users

**\$11**

Revenue per end user / month

**\$5.7586**

ARPU – TCO/user

**MONTHLY ROI**

**110.2%**

(Revenue – TCO) ÷ TCO

**ANNUAL ROI**

**110.2%**

12× monthly projection

**BREAK-EVEN USERS**

**238**

users to cover full TCO

**REVENUE / \$1 INFRA**

**\$2.10**

Revenue efficiency

**ANNUAL REVENUE**

**\$66.000**

12× monthly revenue

## FinOps Architecture Notes & Disclaimers

### 1 – IP Address Planning (CIDR Sizing)

The more redundancy you build into AWS (Multi-AZ, replicas, NAT Gateways, ENIs), the more private IP addresses are consumed. Plan your VPC CIDR block generously (e.g. /16 = 65,536 IPs) and **leave enough address space within each subnet CIDR for auto-scaling headroom**. A /24 subnet provides only 251 usable IPs — exhausted quickly in a multi-tier, auto-scaled architecture.

### 2 – Multi-Tier Architecture in AWS Terms

Separating MACH components (frontend, API gateway, gRPC services, databases) into **different subnets** makes this a **multi-tier architecture** in AWS terms. Each tier needs dedicated subnets with its own Security Groups and NACLs.

### 3 – Intra-VPC Connectivity

Use **VPC Endpoints** (PrivateLink) to access AWS services without internet routing; **Transit Gateway** for cross-VPC connectivity; **VPC Flow Logs** for visibility. Each component in a separate subnet requires explicit route table entries.

\* Outbound (response) data transfer cost uses AWS standard internet egress tiered pricing for US East (N. Virginia): first 10 TB at \$0.09/GB, next 40 TB at \$0.085/GB, next 100 TB at \$0.07/GB, beyond 150 TB at \$0.05/GB. Inbound (request) traffic is free. Effective sizes include TLS (RFC 8446) and JWT (RFC 7519) overhead where applicable. [AWS Pricing Calculator](#)